

LMA Phase 1 Project Submission

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Languages

High-Resource: Telugu | Low-Resource: Nepali

Telugu language Steps

1. Dataset Collection

The dataset compilation process was broadly categorized into two primary approaches: manual data collection and the utilization of pre-existing datasets.

A. Manual Collection

Web scraping and targeted extraction were employed to gather diverse textual data from the following sources:

- **News Articles:** Extracted from popular Telugu news outlets, including Sakshi, NTV, TV9, and Andhrajyothy, ensuring broad coverage across multiple domains.
- **Legislative Transcripts:** Textual data extracted from the Andhra Pradesh Budget 2024 session.
- **Podcast Transcripts:** Collected from popular Telugu podcasts (e.g., *Raw Talks with VK*) to introduce varied domain data.
- **Literary and Religious Texts:** Sourced from Vedapath, Ramayana, and Mahabharata digital files.
- **Bhashaverse (IIITH) Dataset:** Extracted data specifically from the source and target language columns of the Bhashaverse Parquet files.

B. Pre-existing Datasets

For standard pre-existing data, the following corpus was utilized:

- **AI4Bharat Indicorp:** Integrated the primary AI4Bharat Telugu dataset.

2. Dataset Links and Resources

The collected datasets, including both raw and cleaned files, can be accessed via the following repositories:

- **Mahabharata, Vedapath, and Telugu Books:**
<https://drive.google.com/drive/u/1/folders/1daubXuMFe4evJWRo2d5RC1bm9wgs6b0G>
- **News Channels, Podcasts and Cleaned Datasets:**
<https://drive.google.com/drive/u/1/folders/1hDEpu1d0ZSeWseXp1WvsvwrVAd0Kr3VU>
- **Bhashaverse Dataset:**
<https://www.kaggle.com/datasets/tanoojtadepalli/bashaversedatasrt>
- **AI4Bharat Indicorp Dataset:**
<https://huggingface.co/datasets/ai4bharat/IndicCorpV2>

3. Pipeline Implementation

- Collected Telugu data from manual sources and AI4Bharat IndicCorpV2.
- Cleaned the text and converted it into sentence-level data.
- Added M and D source tags and distributed sentences into 256 MD5-based buckets.
- Processed 8 GB of IndicCorp data using streaming.
- Removed exact duplicate sentences within the buckets.
- Created 80/10/10 train, validation, and test splits.
- Trained 5K and 10K Unigram SentencePiece tokenizers.
- Evaluated both tokenizers on 5,000 validation sentences.
- Selected the 10K tokenizer based on the lower UNK rate and better fertility.
- Tokenized the training corpus, producing 863,817,731 tokens.

4. Language Statistics

- Manual sentence candidates collected: 23,272,604
- IndicCorp sentence candidates collected: 46,190,318
- Total sentence candidates collected: 69,462,922
- Duplicate sentences removed: 8,735,800
- Total unique sentences: 60,727,122
- Unique manual sentences: 23,154,617
- Unique IndicCorp sentences: 37,572,505
- Manual word count: 307,468,665
- IndicCorp word count: 364,668,963

- Total word count: 672,137,628

Data Split:

- Training: 48,581,452 sentences (80.00%)
- Validation: 6,069,130 sentences (9.99%)
- Test: 6,076,540 sentences (10.01%)
- Total: 60,727,122 sentences

Tokenizer Statistics:

- 5K vocabulary fertility: 1.675
- 10K vocabulary fertility: 1.441
- 5K UNK rate: 0.0
- 10K UNK rate: 0.0
- Selected vocabulary: 10K
- Final training tokens: 863,817,731

Using the 10K tokenizer fertility of 1.441, the estimated token counts from the source word counts are:

- Manual: $307,468,665 \times 1.441 \approx 443.2$ million tokens
- IndicCorp: $364,668,963 \times 1.441 \approx 525.7$ million tokens
- Combined estimate: ≈ 968.9 million tokens

The actual tokenized training corpus contains 863,817,731 tokens.

Nepali Language Steps

1. Dataset Collection

The dataset compilation process was broadly categorized into two primary approaches: manual data collection and the utilization of pre-existing datasets.

A. Manual Collection

- **News Articles:** Collected Nepali news articles from Setopati, covering topics such as politics, entertainment, and global news.
- **Large Text Corpus:** Collected raw, uncleaned Nepali large text files for further processing.

B. Pre-existing Data

- **Nepali IRIS Research Dataset:** Streamed data from the Nepali IRIS research dataset for corpus construction.

2. Dataset Links and Resources

- **Manually Scraped Data: Setopati News Articles and Large Text Corpus:**
<https://drive.google.com/drive/u/1/folders/1EBgMSGm6tJ03fIfQz2sM9nXuSppZFz-1>
- **Pre-existing Dataset: Nepali Corpus:**
https://huggingface.co/datasets/Tantade/Nepalicorpus/blob/main/nepali_merged_all.json

3. Pipeline Implementation (Nepali)

- Collected Nepali data from manual sources and a Hugging Face Nepali corpus.
- Cleaned the text and converted it into sentence-level data.
- Added M and D source tags and distributed sentences into 256 MD5-based buckets.
- Processed 8 GB of the Hugging Face corpus data using streaming.
- Removed exact duplicate sentences within the buckets.
- Created 80/10/10 train, validation, and test splits.
- Trained 5K and 10K Unigram SentencePiece tokenizers.
- Evaluated both tokenizers on 5,000 validation sentences.
- Selected the 10K tokenizer based on the lower UNK rate and better fertility.
- Tokenized the training corpus, producing 554,121,940 tokens.

4. Language Statistics

- Manual sentence candidates collected: 3,624,895
- Downloaded sentence candidates collected: 30,963,245
- Total sentence candidates collected: 34,588,140
- Duplicate sentences removed: 9,028,670
- Total unique sentences: 25,559,470
- Unique manual sentences: 2,705,965
- Unique downloaded sentences: 22,853,505
- Manual word count: 135,909,429
- Downloaded word count: 367,234,259
- Total word count: 503,143,688

Data Split:

- Training: 20,447,821 sentences (80.00%)

- Validation: 2,554,861 sentences (10.00%)
- Test: 2,556,788 sentences (10.00%)
- Total: 25,559,470 sentences

Tokenizer Statistics:

- 5K vocabulary fertility: 1.579
- 10K vocabulary fertility: 1.382
- 5K UNK rate: 0.0
- 10K UNK rate: 0.0
- Selected vocabulary: 10K
- Final training tokens: 554,121,940

Using the 10K tokenizer fertility of 1.382, the estimated token counts from the source word counts are:

- Manual: $135,909,429 \times 1.382 \approx 187.8$ million tokens
- Downloaded: $367,234,259 \times 1.382 \approx 507.5$ million tokens
- Combined estimate: ≈ 695.3 million tokens

The actual tokenized training corpus contains 554,121,940 tokens.